ActInf Livestream #021.04 ~ John Boik

Livestream | John Boik | 1:54:38

hello and welcome
welcome to actin flab we're in live
stream number 21.04
on april 27th 2021
and today we are in the fourth of a
series of conversations with
john boyk so john it's been really
awesome to get to know about you
and your ideas and work with you on
these jamboards
so as i sort of set things up for a
second and share the jamboard link with
the live chat
go ahead and just re re-catch us up and
then i'll meet you on the first slide in
just a minute
okay sounds good thanks uh so
my name is john boyk i'm a courtesy
faculty at oregon state university
um and the series of discussions that
we've been having is on the
is on a paper a series of papers that i
just published in the journal
um sustainability it's a three-part
series the topic is
well the title is science driven
societal transformation
parts one two and three so uh
we've this is our fourth get together
here and we're just finishing up our
discussion of the last in that series
part three
which which is uh the subtitle design
and um we we covered quite a bit in that
paper in that paper

part three but now today we're going to
focus on just one aspect of that
which is uh the leader framework um
local economic direct democracy
association framework
this is a what i'm i'm terming it a
prototype of design for new societal
systems
and as a reminder to listeners we're
talking about the
the de novo design of new societal
systems
using specifically club model so the
societal systems for
a club the club is a is a small
community
somewhere something larger than say a
thousand
volunteer participants that's how the
in the art the series itself proposes an
r d program
and that's uh the strategy for change in
the r d
program is new systems are tested at the
club level the community level
in cooperation with the global
scientific community who's working on
the technical aspects
then field trials occur at the club
level and if the
if the new systems deliver as promised
and they promise substantial improvement
and uh social and environmental quality
economic
security and various things like that
then the idea is that the club model
would
grow and spread and describe to new

locations
and uh new testing and new trials and
eventually would spread near on a near
global basis
so are we on slide one yet we're ready
on slide one
okay so that's our that's just our uh
our list of
discussions we've had you can check out
the links there for
uh the previous youtube videos
and slide two if you would go there
is uh just where just where to go for
more information so the
the my website for this project is
called principal societiesproject.org
and it uh lists the uh gives links to
the series get some backgrounds
information
um actually has a it on there is a
live or interactive model of the leader
framework
so so part of what will be a big part of
what we'll be talking about today is
actually on the website if you want to
run the model yourself
so they go there for more information
and then
third slide
so today's topic again is the leader
framework
local economic director democracy
association
these are ideas from my 24
2014 book called economic direct
democracy
that book is available for free on the
website or if you

want to spend money you can go to amazon
and buy it there
[Music]
and along with that book i published a
simulation model
of leader framework or at least parts of
the little leader framework
and that is in the international
journal of community currency research
so links are there for that
and again all of these links are
available on the principal societies
project website
okay so now why would i want to discuss
this is
slide 4 why would i want to discuss the
leader
framework in this three-part series a
few reasons one
is that the leader the design of the
leader framework
maybe should say that the leader
framework is intended as an
integrated societal systems architecture
or what i've been calling a cognitive
architecture the the idea behind this
whole series behind this work
is that a society large or small is a
cognitive organism
and its societal systems which for
example
economic systems legal systems
governance systems etc
act as a cognitive architecture so
so when discussing the idea of societal
transformation
my focus here is on uh creating new
societal systems

de novo societal systems design
uh so that they function uh
as cognitive systems they facilitate
societal cognition
and and this is a fitness-based
evidence-based
r d project so the idea would be that
you could assess
how well some did one how well one
design
is doing over another design and
eventually
communities could choose designs that
function very well and are fit for
purpose
all right so the lida framework embodies
the cognitive viewpoint
it also was designed uh to be
implemented via the club model which i
talked about in part two of the series
that's the strategy for change um number
three is
here the reason is that it involves
integrated systems
so uh there's i in the in the series i
talk about six overarching systems
legal governance etc and the the leader
framework uh
integrates most of them and finally
there's a published simulation model
available at least for the
economic some economic aspects of the
leader framework
and at the time and i think that still
might be true this is the first
semi-realistic simulation of a de novo
system a de novo
economic system that i'm aware aware of

number five all right
so i'll get into the we'll talk a lot
about the simulation today and a little
bit about the
you know the the uh structure of the
leader
but let me just offer a few warm-up
thoughts uh
um the
the lita economy is not like what we
face a normal economy
that's it's it's fundamentally different
and in particular it is a
democratic uh it is a cognitive
architecture as i've said and it
is deeply democratic and transparent so
in the in the lead
economy money there is money both the
national currency
in america the dollar and
a local digital currency that i call the
token
so those two currencies
are used as a voting tool in a highly
transparent fair
and deeply democratic process and as we
talked about last time
money is already used as a voting tool
in our
world it's just that it is a
non-transparent non-democratic voting
tool for the most part
there's some people have billions of
votes in the economic system and then
other people have just
you know a small number of votes so uh
the leader framework makes that voting
function explicit and

fair and democratic the uh
the emphasis of the simulation we'll be
talking about in a moment is on the
economic system but
as i mentioned the leader framework
encompasses the
you know all or almost all six
um overarching systems
i i this
this simulation is intended as a
first step and uh it is it's been it
could be made more sophisticated
and uh this so this first one simply
looks at the economic aspects
aspects because that was the simplest to
do for me at the time
um so money is used as a voting tool
and it's and voting occurs in two arenas
it occurs in local markets so i buy
some product from from a local store for
example
local business and it also occurs in
in the financial system what i call the
crowd-based financial system
and we'll talk a lot about that but
essentially
individuals contribute a certain amount
of their
income to the crowd-based financial
system
and then once their money is in the
crowd-based financial system the cbfs
then they have uh they get to choose how
that money is used
so in a sense they get to vote for
they give and offer their votes for how
the community
is what the community wants what the

community wants to fund
how what kind of jobs the community
wants to fund and so on
um so if you had a
if you had an economic system in which
money was used as a voting tool
well uh it would be a fairly
undemocratic
unless incomes and wealth were somehow
equal
so over time the effect of the leader
is to equalize incoming wealth
and i'll show the show the results of
that in a minute the
the latest simulation runs 28 years and
as
you'll see at the end of that 20-year
period everyone who has joined the club
uh is enjoying an equal level of an
equal and high level of income and
wealth
and uh lastly the leader framework
is designed to focus on motivations
that is the any new system developed in
this
r d program the goal is to have that
system be fit for purpose
and as we've talked about the purpose of
a society
the purpose of societal cognition and
the purpose of
i should say the implicit person purpose
of societal systems is to facilitate
societal cognition and to
and thus to ensure
that the organism the society
maintain achieves and maintains a high
degree of vitality

so the any new systems developed should
be fit for purpose
and and if they're fit for purpose
almost by definition they will
facilitate cooperation because
cooperation
and communication is a part of cognition
so the lead of framework and what
rewards barriers to cooperation and with
the idea
that that it makes cooperation so easy
that cooperation becomes the default
behavior
and it also focuses on motivations it
does
not reward people for
uh non-cooperative behavior for example
and that is in contrast to our existing
systems which which i call native
systems in the series
but i in our in native systems
i am for uh
paying my workers a
very long for
the society would reward me greatly for
that and that doesn't occur in the
leader framework
partly because income and wealth are
equalized over time
thus the the part the motivation to
participate in the economic system of
elita or
we'll call it we can also call it a club
model is
not personal gain but community
well-being
well-being of the larger organisms
are the members of a club are you

thinking
uh spatially co-located are they
remotely located yes
locally in space and
it is it was perhaps possible to have a
club that is
distributed over over space
but um i didn't start that way i mean
that
maybe could be something that's added
later but i'm focusing now it's the
easiest thing for me
has been the easiest thing to me for me
to focus on the local market
local area but there's reasons why
the local area is important because many
decisions actually are local
like who will pick up my garbage and
where will that garbage go
is a local decision you think the many
of the decisions many of the
of the our quality of life is largely a
local issue for example
our my economic security is largely a
local issue
and obviously being close in space
allows a greater opportunity for
community building and gathering and
seeing face to face and communicating
communicating face to face and getting
to know each other
and building a deep sense of friendship
and community that way
but yes in the future communities might
be
extended in some way and as a matter of
fact uh the the idea with the leader
framework or the idea with the club

model
is that that the initial focus would be
on developing the
tools and systems for clubs for the
project
and then that would land systems for
club
networks so that
would be by definition non-local
you know non-local engagement because
clubs might be in different areas
of the of the world
okay uh so
is that right yep um
so there's in the bottom left of that
slide
over in the bottom left that slide i'm
listing the six overarching systems that
i talk about in the
in the series and you can
you will see that all but leader so i
would i would imagine
you know that if this were developed
further that health would also be
included but i
just haven't included it yet uh so very
briefly the sec
this uh seven integrated components the
first three
i'm calling uh i call them collectively
the token exchange system again the
token is the
local currency and it and it travels in
parallel with the national currency with
the dollar locally
uh there's a token monetary system that
creates and destroys tokens
there's the crowd-based financial system

which i mentioned already
that uh that to be clear that
cbfs handles both the national currency
and the token
it it issues to local businesses and
non-profits uh it issues donations
interest-free loans and subsidies
and these are all trans transparent
through democratic
decision making and it also has a
nurture arm that
people contribute to and members
contribute to and that
nurture arm provides income for not what
i call not in workforce members
so what when i say that the leader
achieves
income equality over time i mean
for every single a member so
you can think of it as every member
family or however you want to think of
it
but uh regardless of whether that person
is in the workforce not in the workforce
ill not ill employed not employed
in that sense it's it's there's it's a
little bit like a universal income only
it's a very high
income same income that everyone is
enjoying i'll show you that in just a
minute
and then number three in the in the tes
token exchange system is the market
system
so these would be local local markets
that are better members
and uh part of the some of the new
businesses that are developed by the

leader would are called principal
businesses principal businesses and
that's a very special
business model unique to the leader kind
of like a non-profit kind of like a
hybrid between a non-profit and a and a
regular for-profit business
um number four there's a property rights
system that goes along with this
for example a club a leader would have
an ip pool for intellectual property and
the
and the club could also manage other
properties could manage
real estate properties for example or
any kind of property
so it would be democratic decision
making
you know for property management also
there'd be some restrictions on the sale
of principal businesses kind of similar
to non-profits
and that would be high transparency
throughout so the community would know
if the community is using money as a
voting tool the community would need to
know
where their what's happening with their
votes what's
what was the you know what are the
choices that the community is making
with with money and so the flow of money
would be
highly transparent compared to today
um number five is an information and
incentive system
this would include for example education
programs reputation system

the fitness indexes that we've talked
about which allow the community to
understand itself how it's doing what's
going to happen next
uh how how what is the quality of the
design that we're using could we improve
our design
uh and along with fitness indexes the
last one there's computer and other
kinds of forecasting and assessment
methods that would help a community to
understand
what are the trends and how is the
system working and how are we doing
um number six is the collaborative
governance system and that has
three branches administrative
legislative
and judicial so that then covers the
legal system
and finally number seven is purpose and
we've
already talked in each of these
discussions we've talked about the
the importance of purpose and how
purpose purpose follows world view
and um and and
design and metrics follow from purpose
okay so next
okay all right so now we'll
we'll spend most of the rest of this uh
discussion
on the simulation model because i think
that'll
that'll be most that'll make the
description of the leader foreign
and again uh the the
this is this is a first simulation um

it's it's intended to introduce the the
leader framework
um and also just to describe some
general concepts of token dollar
i'll use dollar as the national currency
if you don't mind
token dollar flow and um
and it's the third aim of this uh on the
left
paragraph there is to
demonstrate at least uh you know
conceptually
that income equality can be achieved
um that there's some parameterization of
this model that you some
you can choose parameters such that
uh this goal of income equality can be
achieved at the end of the simulation
the the simulation lasts for about 28
years i believe
um so this is not a this is not uh
you know this is the the the concept
behind the leda is that it
bootstraps its success it starts small
it grows from there
it injects like in the very first year
maybe just a few
members will will join and the leader
injects a small amount of tokens as the
as the local currency into the economy
and makes some very small changes and
then the next year there's
another few people join it makes some
bigger changes the next year makes
another few people join and make some
bigger changes still and the
the growth of the leader occurs
exponentially fast

now i would like i would like to say
that
that the functions of the leader
how it how it works how it achieves
these aims how it achieves income
equality for example
this would all be worked out far in
advance before any
field trial ever occurred so what would
actually happen is
there would be say you know who knows
hundreds or more
of simulations or papers or at least
let's say a large number of simulation
and papers on a variety of topics
would be finished before any community
was ever asked to participate in this
so once that body of knowledge is
developed
then perhaps the the the program would
say okay
what is there are there any communities
in the world
that would be interested in
participating in a field trial
and if you did participate here's the
list of benefits that you might expect
over
if this say a two-year field trial uh
income levels will increase this is
what's going to happen
this is this is how participation will
increase
this this is you know this the the
trajectory would
be laid out through a variety of
simulations
so everyone should know what they're

getting involved in what they expect
from this and
and there should be no surprises from
the field trials
now that doesn't mean that the community
has no choice in their behaviors and
what they do with the system they
the community drives the system but the
system is designed such to achieve its
purpose right fit for purpose
um all right so um
um on the left there i just want to
emphasize this
that this simulation conveys broad low
level low resolution design and
intentions for an actual
token exchange system so i'm not
predicting that a token exchange system
would work like this
this is rather a
illustration of the concept of the token
exchange system in the leader framework
and obviously this simulation
can and hopefully will be followed by a
large number of other simulations that
are progressively more sophisticated
this is a in many respects a fairly
simple
simulation and there is tremendous room
to
expand it and improve it
um but it's still useful and in
particular that's because i
made the simulations stock flow
consistent which means that
no no monies come out of thin air it's
all
you know if money leaves one agent then

another agent
receives it so all the flows of money
are accounted for
and just that alone allows one to say
that okay logically
you know logic mathematically this all
adds
up um and i would
lastly i would say that this is a what i
call a semi-realistic simulation
that is uh income level starting income
levels resemble real income levels tax
levels tax rates resemble real tax rates
[Music]
thus the initial conditions are some
silly realistic
but but again the simulation is very
simple so i start with
a semi-realistic initial setting and
then
certain things don't change over the
years it's just sort of
the only thing that actually changes is
just what is pertains to the lead
itself the members themselves so if you
don't join and meet
in this in the simulation if you don't
join the leader over the period of years
then your income stays just what it was
and um
that nothing really changed just for you
you keep the same job and
keep the same family and things just
move ahead
all right yeah please
okay you could address it uh now or
later but
what is the relationship between your

simulation and game theory
and what is the relationship between
your simulation and active inference
oh okay um
okay so um in a this is an agent-based
model
so in that means uh you know
between the lines that means that there
are individual agents that make
decisions
of what they want to do and that
determines how the
the history of the simulation goes
um in a in a highly sophisticated
agent-based models the agents themselves
might be very complex
and they might make decisions about
complex situations
and since this is a just a simple
introductory model
the agents are relatively simple um
like for example a family and they make
fairly simple decisions
and the main decision they make
in this uh simulation is whether or not
to become members
in any given year does a family want to
become a member
secondly they'll make a decision on how
they what
they choose a wage option which is a
construct that i'll talk about in a
minute
so there's not very many decisions that
uh that individuals will make here
so it is like a game theory but it's a
very simple game
and later simulations could be made much

more sophisticated
and in relation to active inference i
obviously i i might say that i created
these
the book and the simulation model long
before i
was aware of what active inference was
so i what the book neither the book nor
the simulation model
mentions the cognitive aspects and that
came a little bit later
in my in my development and as the
project developed in my mind
but but one could
easily imagine how active influence
inference might fit into this
for example you know as i said a a
community
is considered to be a organism
it's understood as an organism a
cognitive organism
so just like every organism it has to
fulfill its
intrinsic purpose which is to maintain
and achieve
vitality and it and it
in the active inference scheme it um it
assess it it receives information from
its world
through its senses it updates its models
it chooses actions
that changes the world and then it
receives new sensations based on the
changed world and perhaps updates its
model
further or chooses a different set of
actions or continues with the same
action

but but that's how active inference
works for an organism but that's how
active inference would work for a
community too
the community senses information it
decides you know it learns what it can
learn from the current uh
sensations it makes decisions about what
it'll do next
in a variety of ways what will it do
with its solid waste
where how will it generate energy what
will it fund next
uh what what businesses will it fund
what nonprofits will it fund what is it
what is its targets does it adjust its
targets
so these are all decisions that uh that
the
organism of a community would make as it
tries to
to achieve and maintain vitality and
fortunately uh
since this is a fitness-based
evidence-based
program the community would know at
every moment
how it's doing what it's what its
indexes are it's you know it's indexes
for vitality for example or it's or it's
indexes for uncertainty about the future
so as it tries as the community tries to
reduce uncertainty about its
essential variables then you know it
would have an immediate
feedback as far as like scores about how
it's doing and
those scores and that information could

help it then to update its models
or uh and choose the same or different
actions
and so that's the that's how active
inference might fit in in
the community as a whole but you could
imagine that any
you could focus in on any individual
part of this stay just a principal
business
and the principal businesses the
principal business could also be
thinking about its cognition
in terms of active inference it too
gathers information
and then makes uh makes adjustments and
learns and
chooses actions and
another example is the cloud-based
financial system itself this is a
decision-making
organization structure and that whole
structure
could be modeled using ideas from active
inference
cool it's interesting how you developed
the idea
and applied basically the evolution of
computational
modeling to it and that helped a few
things maybe come out into
relief like that's that multi-scale
aspect and then also
the fact that you're pursuing not the
maximum gdp
you know not the the total divided by
the number of people
which you're maybe going to get a higher

average with one billionaire and
everyone else has zero
but that's not the pursuit it's actually
a
uncertainty minimizing trajectory policy
sequence
that's starting to sound a lot like
active inference so it's kind of cool
how it dovetailed together
right right this is these are these are
you know these are probabilistic this
you could say the community is
developing a probabilistic model of what
will happen next what will happen to me
and uh and that fits in very well with
the active inference scheme and
individuals in the community are
developing a generative model of
of how i see the world and what will
happen next
and as a matter of fact since the that
since it's so important since
anticipation is so
important in cognition the individuals
in the community individual
organizations such as businesses and the
community as a whole
would be developing anticipatory models
whether those are
formal models uh you know in computers
or whether they're informal models in
their
in their minds about how how i see the
world and how i see
what will happen next if a b and c
happens
so all of that information that full
anticipatory

capacity of individuals and
organizations and the community as a
whole
can be taken taken into account in the
leader can be used in the leader for
the for its own good
all right number uh slide eight yep
we're there okay all right so um
this uh this image on the right
really captures much of the much of the
simulation
and much of the idea of the leader as
far as the token exchange system goes
um and before i
describe it i just want to emphasize
once again that the token exchange
system is just one part of the leader
and there's other parts that are not up
not in this simulation at all for
example the governance
governance system is not in the
simulation at all
so this is just flows of currency and
this
uh figure one on the right describes how
that works
to to simplify this a little bit
let's just look at the arrows in blue
and the arrows in blue are both tokens
and dollars that flow together
and we could start at organizations
these are the agents of the model
organizing the five agents organizations
persons
cbfs rest of counties which is the rest
of the world
and government which is local government
um just looking at the blue lines we see

there's kind of a circle being made
between organizations persons cb
cbfs and then again to organizations and
that's
essentially how how
how how things flow organizations
persons are employed at organizations
and organizations pay wages to persons
persons spend money back into
organizations when they buy
goods and services persons support the
cbfs with contributions
and the cbf fs supports persons
in particular not in workforce persons
with what i call nurture support
the cbs
also provides loans subsidies and
donations to organizations
and the organizations pay back the loan
principal i might add that the the
concept of loans here
is uh is the token itself is
designed to be a non-inflationary uh uh
um currency
and loans are offered interest-free
so what you don't see here in this
circle i just
described what you don't see is
investment for profit
that does not occur in the system um uh
especially with relation to principal
businesses it doesn't occur and
obviously doesn't occur with non-profits
because
non-profits are by definition not for
profit
so so if i'm a person i receive
wages from my employer or maybe i'm

self-employed i sell things to the other
persons
and i have to contribute a certain
fraction of my
income to the cbfs and once it's in the
cbfs
i can organizations organizations will
apply to the cpfs for loans or subsidies
or
donations and
i along with my fellow colleagues in the
cbfs
i have a pool i have an account there
and i can decide you know i like i like
that idea that this new business wants
to do they're going to make a new light
bulb or they're going to make a
this is funding for a small organic farm
or whatever the situation might be
i might like what they're doing and i
might see value in it for the community
again this is the investments are
not related to profit so i'm not i'm not
thinking the way
i would think in today's world in the
native economy if i have money
in the native economy i'm thinking where
can i put my money that will return to
me the greatest
gain personal you know my personal gain
and finances
in this system i'm asking
since since personal profit is not part
of the picture
i'm asking where can i put my money that
will receive the greatest gain
in terms of fitness scores and community
well-being and quality of life for

myself and my children
so those are the kinds of topics that
become pertinent then
when when the personal profit motive is
taken out
so um so
again in relation to the native economy
the native
the government the say the us government
also makes subsidies to various
businesses
but here all these choices
are deeply democratic you know
given that incomes are more or less
equal everyone has more or less
the same amount of money in the cbfs
same amount of currency
so everyone has a is essentially equal
say in which which businesses get
funded which businesses what you know
whether they get subsidies
or whether they get donations or whether
they get loans
and the reason for subsidies rather than
loans and donations might be that um
well that for one thing you have to pay
back a loan so
if you if a community wanted to support
say a bakery
and they thought that um you know this
bakery is really great they don't
they're they're not making a lot of
money so they're not going to be able to
really pay back loans very well but
they're
maybe even they're operating at a loss
but there's just such a great community
resource and we just love this little

bakery so much
that we would like to give this bakery a
subsidy
so that it can continue to operate even
though it's losing money every
year you know that would be one example
of a subsidy or there might be we want
to subsidize a hospital
or we want to subsidize any any
operation that perhaps
is not actually making money and
couldn't pay back a loan on the other
hand maybe
some business is doing well and we would
be willing to support it but we want to
give it a loan instead of a
subsidy donations really apply more to
nonprofits and subsidies really apply
more to
businesses coach give a thought on that
john all right
yeah yeah it's like if uh
a person who's granting money has ten
thousand dollars
and they're deciding whether they want
to be the one who gets to decide where
the 10 000 go
or whether they're going to give it to
the cafe
who's going to spend it potentially at
other local businesses
and allocate it a different way and the
employees are already taken care of
and it's going to contribute to public
good especially a locally
um defined one and so it's like yeah
i'll let them spend my ten thousand
dollars that i was

just going to give away anyway right
right right yeah so so
that raises a really a good point that i
wanted to make is that
what's really happening here
even though it's currency that's
circulating what's really happening here
is just it's just a way to ensure that
everyone gets the same amount of votes
in this economic arena and what people
are voting for
is not so much money if money isn't the
resource the
this as especially in a mature leader
the money is circulating
as all as i'll show later so it's so
it's not really a shortage of money like
how will we get the money for this or
where will the money come from it's not
that it's using money as a voting tool
to allocate
other resources like if i give
a um if i support a bakery say with
with a loan or a subsidy
what i'm saying is i'm i'm voting
for human resources of the community to
be spent
you know in this bakery i want this
bakery to employ
these people and i want the bakery to
provide this service to the community
right so i'm voting for
the service and the allocation of
resources
that are embodied in this bakery or it
could be a hospital or it could be a
could be anything but
that's what we're really voting for is

how we spend our time
what do we what do the humans and this
what are the members
in their in their life and their 24
hours a day and seven days a week or
whatever
in their year how do they spend their
time how do we want to spend our own
time
what jobs do we want to work in as a
community
how do we want to spend our time do we
want to be employed by non-profits do we
want to be employed by
regular businesses do we want to be
employed by principal businesses
this is a the evolution of this economy
is the is the decision-making process of
a community to decide
how it wants to spend its time and where
it wants to put its focus
and how how what kind of conditions it
wants to operate in
okay all right um
so that's a little bit of an
introduction to the flow of
tokens and dollars together um
uh maybe i will say one more thing
before we get off this is that
suppose that um and this might be
obvious from things i've already said
but i just want to emphasize it but
suppose that
me suppose that me and a group of people
in the cbfs decide like hey it would be
really cool to have a
bicycle manufacturer in our town so
so together you know the the 12 of us or

the
50 of us or the 100 of us decide like
yeah let's support this bicycle business
now every it's natural that not all
businesses are going to
make it you know some are going to fail
and that's natural and good a community
would want to take some choice some
chances
and fund some things that maybe even you
know the reasonable chance it will fail
but
if it doesn't fail it would be great so
so we fund a bicycle business and it
fails after a few years
okay fine the same money is still
circulating in the community nothing's
really changed
now with what has changed however is the
the allocation of human resources that
were allocated to the bicycle business
now can be allocated someplace else
because the bicycle business doesn't
exist
and in the meantime those employees
might become then
not in workforce members right they
maybe don't have a job for a short
period
so they go from the unemployed category
in the persons or
person's box to the unemployed and not
in workforce box
and fine but you know their income
doesn't really change
so they were you know
they're not out in the street and
neither is the business owner the guy

who did the whole thing he's not out in
the street he's
his income hasn't really changed too
much especially if he's a principal
business
so he's fine the employees are fine
everybody's taken care of
the community took a chance they said
hey this would be a good experiment
let's have a bicycle
shop it didn't work out but now we'll
just re-allocate
those resources to something else some
other decision
so you know the the funny that when i
try to explain this to people
the the and i'm using terms money and
currency
it's really it's difficult for some
folks to get past
their interpretation their the common
interpretation of what money and
investment and
you know currency means and uh this is
as i said earlier this is a lita economy
but it's not like any kind of economy
that
we're we we think of this is really just
a decision this is a cognitive system in
short this is the description of a
cognitive system that a certain flow of
that
is uses what i call currency
if i can make a point on that john it's
like if you use the metaphor
of money a lot of times people debate
uh characteristics like fungibility or
store value or transfer value

but if we think organismally and we
think about let's say
blood flow or neural information or
hormonal signaling
like the goal of the muscle cell is not
to accumulate
insulin it's to use it as a cue right to
know when to change
and so if you think about inflammation
flow blood flow no cell needs to hang on
to the red blood cells and capture
things around it because
it's part of a bigger plan so i think
that that's a really nice
attribute you captured yeah excellent
analogy and that's very much what the
leader framework is like it's kind of
like a body
kind of like an organism that that
requires the different cells in that
organism to receive
blood and food and information and other
things like that
so and also on that point when i say
money is used as a voting tool
even more broadly it's used as an
information tool
so the community is exchanging
information when they exchange money
uh uh still on slide eight uh i've
explained now the blue lines the token
and dollar flows
but i haven't explained the green lines
so
uh this this
the new club in this model
uh as in as in all models for new clubs
the community

remains part of the larger world it's
there's no the club is not segregated
from the rest of the world
uh apart apart from the rest of the
world it is engaging in all the normal
stuff it's paying taxes to the
government
it's purchasing products from member
member organizations but also from
non-member organizations and amazon.com
or you know whoever
so uh you know it's it's participating
in the normal way and this lead of
framework or the club framework is
another layer
of organization on top of the normal
world
um so these green lines are showing how
dollars are flowing
and again this is a simplification of
the real world but
it gives the idea of still so persons uh
are
paying the dollars in taxes paying their
taxes and dollars i should say
and the government is also giving some
income support to unemployed persons
that
also happens uh the government is giving
dollars
in the form of grants and subsidies and
contracts to local organizations
and local organizations uh i and i
created this as a kind of a
simplification but
local organizations are exchanging goods
and services with the rest of the world
so dollars

leave the community when an org local
organization for example
purchases supplies from some far distant
business
and at the same time
[Music]
organizations outside of the club
outside of the county
are are purchasing things from the
member organizations too
so uh yeah so that's
that's a simplified version of how
dollars are flowing
and obviously now we got a problem
because uh
you know we have our first challenge
maybe you could say one challenge is to
somehow design this system
so to reach the aiming number three
there that income
is becomes more equal over time but
now we have the problem we've created
this this local currency the token
and we have to pay taxes on income to
the government
and we can't pay taxes and tokens the
government's not going to take it
so we got an immediate challenge of like
how do we
make this thing work so that every
individual
has enough dollars to pay its
the taxes it needs to pay and
organizations have enough dollars so
they can purchase
you know goods outside the county in
dollars
and yet and yet the whole thing can grow

and
and and reach its the desired aims so
how does that happen
another question designed the question
is
how do you make this thing good enough
that anyone wants to
join like fine i've drawn drawn this
picture here of
you know conceptually how it works but
if i if i
don't receive anything materially from
joining why would i join um
now there in actuality there might be
lots of soft reasons to join i might
feel like i'm more a part of the
community or i might feel like my voice
is heard more or i might feel
that uh you know by joining we're able
to
make better decisions together about
things but
in this simulation i wanted something a
little more
substantial than that so
so i in the simulation i made the rule
that a person will join a family will
join this
if it increases their income and they
won't join
if their income doesn't go up so
uh that's that that becomes the
challenge then how do you inject
enough tokens and get a good enough
circulation of tokens and dollars in the
in this
membership such that
my my income goes up from joining or

at least most people's income everyone
that joins their income goes up
and uh how then how many people have to
join to make this thing work
like two percent of the population in 30
years or 50
percent of the population or 90 of the
population like what what does it take
so these are all the questions that
we're going to dive into next
uh uh first a few assumptions just to
explain a little bit how this whole
thing works uh the not in workforce
population is static i think it's
whatever it is 30 36 or something like
that you know there's just
a big as in the normal world a big chunk
of people
don't participate in the workforce
they're either they're
they're disabled or they're staying home
they just stay home and don't work or
whatever reason um uh
i chose a seven percent uh let's see i
think
i think the unemployment rate starts at
about seven percent
something like that um uh everyone in
the county
is uh uh adults and grouped into
two-person families just for simplicity
so household income and family and
computers anonymous
the purchasing power of the token is
taken to be equal to that of the dollar
so when you hear token
think value of about a dollar
and uh structural unemployment is one

percent so
you know you can't get much below that
no matter what you do
and uh all employed persons work
full-time you know just a few
assumptions to make this simulation
simple
and again later some simulations could
be more complicated
all right give a thought there john yeah
uh there's almost two extremes for the
token
one is actually something that was tried
in my little town of davis
california with the davis dollar which
was a one-to-one
u.s dollar it was a stable coin before
there was stable coins
and so then they pegged the local
currency it didn't end up continuing um
it's not active to this day but it was a
pegged
coin and then the other alternative
would be like
a token that essentially people could
they could make an nft out of it or they
could
go full crypto and they could inflate it
and speculate on it
do all sorts of leveraging if it was
really being traded on the open market
so there's kind of a continuum
between a total protected in-group
currency something that's pegged to a
national currency
all these other options for the
tokenomics so it's really
um it's it's an interesting starting

point but i hope others
when they're viewing this see that
there's a lot of space for
experimentation
yeah there's great points and um
yes there's a tremendous amount of space
for experimentation
um and and on that note i want to
emphasize that i'm thinking of this as
kind of a prototype concept
is just to generate ideas just to give
an example
the goal of the of the r d
program if it were funded is that
it would it would evaluate and help to
evaluate evaluate facilitate the
evaluation of any number of
designs and any number of design
components that maybe are
interchangeable or can be mixed and
matched
so this is just one concept and i would
hope that
over a period of years many concepts
are um are tested you know
obviously tested first in simulations
and in
in bench scale studies and things like
that and then if the ideas are really
good
then also in field trials in real
communities
now having said that though i do want to
emphasize that
again this is a science-based
evidence-driven
program so so it's not just
a question of hey what do you guys want

to do well we want to do this and what
do you guys want to do well we want to
do this
there you actually have to show that
your design
is good or better than others or how it
can at least you
at least describe how this design
compares to another design
in terms of the fitness metrics in terms
of fulfilling the purpose
of a system so there's a general general
agreed upon
a concept of system fitness and system
and fitness metrics uh how to there's a
generally agreed upon
uh path to assess a system and then any
design that is
you know that is uh that is good
relative to to the
fitness metrics and and fulfilling its
purpose then
you know this is that's the idea like
any conceivable
design that improves fitness is a good
one
right that's one point now you also just
on the currency i wasn't actually going
to talk about this but
like you brought it up so here we go you
i think you mentioned
two kinds of currencies one is kind of a
potentially speculative currency like
bitcoin
another is a currency backed by the say
the us dollar or something you know like
you have everyone has to
put in a dollar and then they get a

dollar out of
local currency my concept for the leader
is neither of those so uh the
the concept here is that the leader is
um
there's a word for it it's escaping me
right now um
it is just simply created out of thin
air this is it's an accounting system
so in the same crypto thank you
also crypto is by fiat too it's by
computational fiat
yeah yeah so this is by no computation
though
this is just fiat but because we need
more current we
need more tokens so this is an
accounting system
and obviously there's a limit to how
much tokens you create because the
the concept here is that the token holds
the value of about a dollar
so if you created 10 billion tokens and
flooded a community on day one
the the token would be worthless right
it can only be created to the degree
that it can be engaged with
and employed in the circulation it's all
about circulation
and in particular it's not speculative
so you cannot sell
in this system you cannot sell your
tokens to anyone else
that's because i in this system i wanted
people to be
uh you know engaged
you know this is like a voting system i
wanted people to be engaged in the

voting
or community choices and it just
wouldn't be right if i sold my
votes to some other third party and gave
them a whole bunch
that just doesn't mean i can't uh uh
entrust like in the cbfs for example
there might be a there might be a
community organization that says
like an investment group in a sense a
social investment group that says hey
we've been watching you know we pay a
lot of attention we
we model the success of businesses and
the needs of the community and
if you entrust tokens to us and dollars
to us we will
choose good investments for them that we
feel are the best
and you know improve the community well
you can do that that's fine you know
that's all great
but you can't sell your tokens to
anybody
and obviously then they're not backed by
any resource dollar or
otherwise but that's just that's just
this design so
as you said you know other designs would
work differently and you could model out
which ones improve the fitness of the
community uh best
both current fitness and anticipated
fitness
all right number slide 10. yep
um so some additional notes
um as i s as i said earlier one of the
aims of this

of the simulation was to show away a
logical way
in which incomes can grow and equalize
over time
but another way to say that is that
elita essentially pays people to become
members i mean that's how the simulation
works if their income goes up
by becoming a member then they become a
member
um so you know it's that's a that's a
good selling point
if if if elita has started up in my
community
and i realize that by joining my income
goes up
well you know that's a good reason why i
might want to join
um and another
thing is that we'll talk about three
constructs here that's kind of the
machinery that makes this
whole thing go it's the income target
the token share of income
and the wage options so we'll be
describing those shortly here
um i already mentioned that a leader can
create as many tokens as it can
productively
use limited by inflation concerns
the simulation goes for 28 years the
first 15 of that i term the growth
period
and this is when the number of
participants in the community
is rising the number of tokens generated
by the community is rising
uh the the club itself the leader is

expanding
in in all ways and after it reaches its
its peak at about 15 years and as it
turns out
that's the 90 percent of the local
population end up joining the leader
again because
their income goes up um and then the
following years the 13 years following
that are called called the post-growth
period
and that's just equalization period
where everything kind of settles out in
time and reaches a kind of a
homeostasis point
um
uh let's see here oh uh and on the right
with regards to the leader we can talk
about
family income pre and post cbfs
right so so if you're if you need if
you're
required to contribute a certain
fraction of your income to the cbs
well then obviously your pre cbfs income
will be higher than your post cbf
cb fs income uh when i do when i do
comparisons
here i'm really making comparisons on
the post
cbfs income so you can think of that as
really pre-tax uh take-home income
all right um and maybe one last thing to
talk about
oh go ahead one funny note on that is
depending on the uh
governing tax body and how they regulate
whether a certain kind of crypto

transaction is taxable or not
every transaction of a token could be
considered taxable
i'm not a lawyer obviously and this is
the future but i'm just saying that
depending on
what kind of overarching tax structure
exists
it could completely make or break this
total structure for making it tax-free
to making it something that grinds to a
halt
oh if if uh if this was tax free the
token was tax-free if i could get income
as tokens tax-free
then this whole system is going to work
like a charm i would say
i've designed it under the assumption
that everyone will have to pay taxes on
the tokens that they receive
so uh that's the more conservative
assumption and
it and it holds true and we'll show how
this how it all works out
um i do want to uh emphasize
that this that the you know the
the the kind of the structure of the
leader
and the way the cbfs works and the
contributions to it and things like that
it's not dependent on the good will of
members
these are the it's these all occur
because of the rules of the club so
if you want to become a member here are
the rules the rule is
you know the you know there's a variety
of rules that you follow and once some

of those rules
are encode what your contributions to
the cbfs will be
again once you put the money in there
then you get to choose
you know largely you get to choose how
that money is spent
all right on to uh eli
yep so i mentioned that there's uh three
constructs that sort of drive this whole
thing
and the first is the income target i
also mentioned
that simulations would be available long
before
the first field trial ever ever occurs
so in these simulations
the community that's being simulated
would choose an income target
and that income target is a is a
increasing
level of family income up until
up to a certain plateau
and it levels up so that path
the expected path of that increase in
income family income is part of the
design of the leader and it's
and different communities could choose
slightly different uh
uh um functions or paths to get there
but that's the concept that that
everyone knows in advance
how we're gonna get to this uh plateau
level you know how many years it'll take
at what rate
for example like that um
a second uh construct is the wage option
and uh this is to address a certain

problem
so for example i belong i'm in a family
my husband and wife team
and i make a lot of money and my wife
is not working say as an example well
then she could join the leader
and be included in the not in the
workforce group and she could get a high
income just from being a member of the
leader
but i'm going to stay outside the leader
because i'm making a ton of money anyway
and i don't need the leader you know so
the
best choice for us would be i i stay out
of the leader she joins the leader and
our income goes up considerably because
of it but that would be a little bit of
a freeloading
uh situation and would not be good you
know it wouldn't be good for the lead as
a whole so
to prevent that kind of thing there's
two wage options
if our family income is of a is
as this uh income target rises in any
given year
uh if our income target is
above it or if our income say i'm
working in some business if our income
is above the current year target
then um uh then we choose wage option
one
and the and and the club
will give us a stipend you know above
our normal income in tokens a small
stipend so we get a little bit of
we're doing well we have a lot of money

because our income is high
and we join and we get a small amount of
tokens that we can play with in the cbfs
and stuff
on the other hand if we our family has a
low income
lower than the target and we join then
we receive
the whatever current year income target
is so in the first year that might be
low but each year it increases
uh monotonically and by the time the
simulation is over
everyone essentially is choosing this
wage option number two
and is receiving the um receiving the
current year income target at its
plateau
that's a little bit about that and then
um
[Music]
i i should say that the uh i the
the set of families whose initial income
is is is below the 90th percentile
that i consider those the target
population and as we'll see
the leader uh increases
incomes for 90 of the population and
it's that 90
that joins the leader by the end of the
simulation
and there's also a token share of income
target tsi target
that is in any given year um you know
how are the dollars and tokens
uh you know what is like if i'm going to
purchase
goods and by local community and local

businesses
how many tokens am i going to give
relative to dollars
that's also a design parameter
okay slide 12.
one question is you mentioned a
two-person household
where one person had a high income and
one person didn't um
and then you said it'd be free writing
if one person joined and the other
didn't but
what about one person wanting to join
who doesn't have an income
i mean where does that play in
oh oh oh that didn't play into this
simulation because for simplicity i
assume that everyone was in a two-person
family obviously in a more sophisticated
simulation you would have young adults
that are single
and you know and so on so that the
simulation just
doesn't have that level of
sophistication but i would i would argue
though that
i'm showing by the simulation the main
concepts
and that the main concepts make
mathematical sense
so i i don't think it would be all that
difficult to have a bit more
sophisticated simulation where there's
actually one-person families and things
like that
but good point um okay so i'm gonna in a
moment i'm gonna show you some of the
results of the simulation and

just just keep in mind that that's a
simulation for a hundred thousand people
and i'm using uh for initial income
values and income distribution i'm using
data u.s census data for 2011
from for lane county oregon from the u.s
census
so that's that's how i'm making this
semi-realistic in part
is i'm trying to mimic the distribution
that was in lane county oregon in
2011 and then and then that's to start
with
year zero and then every year it changes
as it changes
and if you never become a member then
your income stays at the
same level not obviously not in
not considering inflation in this model
it's just a simple model
and so we're kind of explaining the
concept of inflation for the moment
and it could be easily added in a more
sophisticated
simulation um uh
as far as other ways to make this
semi-realistic i as
at the bottom of the left-hand side
there i choose a tax rate that is a
reasonable tax rate
there's a bunch of other uh
parameters that i chose to be reasonable
parameters of what i could
you know what's a reasonable tax rate
what's a reasonable tax rate oh i chose
uh
for individuals i chose ninety percent
of adjusted gross income you know and

then
everyone gets a 2500 deduction and you
know things like that
so like in the ballpark what that's what
i'm arguing is that these things are in
the ballpark
um and again a more sophisticated
simulation could look at a broken
you know like a more complicated tax
rate or something
um i also look for example like you know
what
what appears to be the average rate that
people
donate to nonprofits just in dollars
just regularly and
you know i chose a percentage of income
for that too things like that just to
just to kind of get semi-realistic and
it turns out that the simulation is not
very very sensitive to those
parameters so they could be changed a
bit and the same the simulation should
show the same results
now on the other hand there's some token
exchange system specific
parameters and that they definitely
drive how this whole thing works
there's the uh some of them i mentioned
for the constructs there's the
the formula is how how uh the
income target rises every year uh that
makes a big difference and
and so to some of the others the the
amount that people contribute to the
cbfs makes a difference
and so on uh and speaking of a month
they contributed to the cbfs there's

different
kind of categories of the cbfs there's
the nurture category for example the
subsidy category the loan category and
so on
and there's what i call ear marks for
all of those in this simulation
like a certain percent of income goes to
subsidy
group and cpfs certain percent goes to
lending
a certain percent goes to nurturing et
cetera
i called it nurturing by the way is
because that's i wanted to convey the
idea of
the community is nurturing itself
through this process
it's it's providing votes to those
people that wouldn't otherwise receive
votes
from like from their employer so again
money is a voting tool
uh okay all right initial employment
rate is seven percent unemployment rate
is seven percent
uh and the initial fraction of employees
who work in non-profit organizations is
about seven percent
now uh when uh when uh when a community
say the leader existed and the community
was getting together and
kind of form forming to use elita they
would have to make all kinds of decision
decisions
like what kind of income target function
are we going to use how many years do we
think

it'll take to reach our you know full
capacity
uh there would be a whole bunch of
decisions like that to make
and uh one of them would be um how many
what kind of
makeup of the our like makeup of the of
the
organizations do we want do should we
all work for nonprofits like what if we
what if a hundred what if 90 of us the
membership actually worked for a
non-profit
that's kind of an interesting question
you know i and i would argue that this
will work
the community could choose like fairly
extreme values and
should work as long as the system
continues to function
as a cognitive system and as long as the
community is using it to solve their
actual problem
so if the community wanted to to have
most everyone work in the nonprofit
sector and wanted the nonprofit sector
to
grow food for the community and provide
services to the community
it could do that i think it could do
that if it
if it instead wanted to fund principal
businesses to a large degree
and nonprofits to a smaller degree it
could do that so
this again this is a this decision
making process is a way for us
for a community to decide what it wants

to be
what flavor it wants its economy to be
where where it wants its individuals to
work
like you know being a member of a
community like this i would want to
choose jobs that i that are useful to me
and to others and that i would want to
work in you know like
whatever whatever those would be but an
interesting thing though is if if a
community
chose to fund a large part of its
workforce in non-profit sectors
it could have say you know a large
number of biologists out of repairing
forests and street and streams and
you know it could have a lot of forest
workers uh doing
you know repairing things there could be
a substantial amount of
ecological reject narration
funded from this but there could be
other things it's whatever the community
feels
is going to increase its fitness you
know it's its capacity its well-being
its quality of life the community can
choose not only which jobs to fund but
what kinds of jobs what kinds of
organizations
all right all right uh on to the next
slide 13. so this is the way the
simulation occurs and it goes
uh in annual steps and in each
in each zone each year the first step
as new members are added according to
this

uh expected participation rate and uh
and some of those people who are already
working in what i call lead of funded
new jobs
l f n j uh that is you know the lead of
funds an organization but
that organization might go bust and then
those people would lose their lead of
funded new jobs
and they switch over then to the to the
unemployed or not in a workforce
category so uh
number two every year some people lose
their jobs some members lose their jobs
uh the number three the cbfs makes
nurture payments to those people who are
not in the workforce who are unemployed
um uh
i'm just gonna i'm not gonna go through
each of these
but i will not mention number five um
so the leader has injected tokens into
the economy you can't pay taxes with
tokens
so so by definition some dollars have
to come from elsewhere another
factor is that in a poor community it
may have a shortage of dollars
relative to the rest of the world right
other other
other areas in the country might have a
higher percentage of dollars in
circulation
so somehow this club this leader
has to pick up dollars from elsewhere in
order to fund its uh you know in order
to achieve
the level of income that it's aiming to

achieve
and it does that by
one main one main
process and that's by
having a kind of a sophisticated buy
local program
that is uh it's the
community is supporting local businesses
and um spending a good chunk of its
of its resources in the local businesses
and then by definition that money is not
flowing out of the community it's
staying in the community longer it's
circulating in the community longer
and uh it the community obtains dollars
from elsewhere by uh by having it
by adjusting its trade balance with the
rest of the world
so one of the one of the the
the statistics that the community would
be tracking is what is our trade balance
with the rest of the world how are we
doing are we
are we adjusting our trade balance as
necessary to bring in the dollars that
we
want and uh in this simulation
as you'll see the the the families they
have the
families actually reach what would be
considered very high income from this
the 90th 90th percentile of starting
income
so they families would receive quite a
very high income
and that would include tokens and
dollars and as it turns out
the dollar portion of that is

about the national average if you looked
at all all incomes in the u.s
that community ends up in a sense
obtaining its fair share of dollars
so if every community in the country
were to do this
there wouldn't be a shortage of dollars
because those dollars would just be more
equally dispersed everywhere
and that's kind of important you
wouldn't what you would not want to do
inside
d program is to design a system so that
a
small community can do exceptionally
well at the expense of everybody else
so that's that's not what this is
um so uh number five was that you adjust
the trade balance as part of these steps
every year
and again this is a very simple
simulation so at the end of each year
all the individuals spend whatever has
gone into the cbfs
or to taxes or for other purposes like
that they spend that in local
organizations
and finally number seven individuals may
pay taxes to governments in dollars
yeah so that's it so that's kind of the
steps you get there that's how it sort
of works
mechanically here's some of the uh
parameters that i used
over on the left are the kind of general
parameters
uh excuse me those are the the targets
for the for the community so the income

targets both starting and ending values
uh and this is for the growth period
which is a 15-year period
so the uh
income target starts low at the
essentially at a
what would might be called a minimum
wage right
and then it rises up to uh
over almost 110 000 tokens and dollars
so
110 000 family income in currency
and that is the 90th percentile of
starting values
so by the way i've already said that
every year the
the club essentially pays people to join
so uh obviously in year one it's not
asking anyone to
reduce their income down to the minimum
wage right so
no one is asked to to lose income in
this
uh people join because their income
increases
now some people might join because they
just want to be a part of the
community part of the leader and their
income would join
their income would increase too if
they're above the income target that
year
their their income would increase by a
small stipend paid in token
so even if i'm making a million dollars
i can join and i can receive some tokens
and i can participate in cbfs and
other things like that um

[Music]
so that was the table on the left and
the table on the right are some of the
ear marks that i
use in this particular simulation and
you might notice that the highest ear
mark
there is for the nurture arm so a good a
good chunk of my
kind of raw revenue come in my raw
income
a good chunk of that almost 40 percent
is going to the nurture arm
of the cbfs and that's pain again for
anyone who's uh unemployed or not in the
workforce
and uh other than that so some of my
income goes to the lending arm some goes
to principal businesses some goes to
nonprofits
goes to subsidies of various kinds goes
to donations for nonprofits
and uh and so on and i also have
obviously if i'm putting money into this
lending function
then i'm building up a kind of a savings
account in the cbfs
that if people are paying back my loans
then every year i'm maintaining a a good
chunk of money in the cpfs itself kind
of like a savings account
i think uh yeah that's about it there's
a couple of rules of thumb that i
realized in trying to find
uh parameters that actually make this
thing go and the
one rule of thumb is the the nurture arm
has to

have a high enough air mark so that that
actually functions as as it's supposed
to
and uh and also the donations earmark
for
local nonprofits has to be high enough
so that any non-profit that i fund can
actually pay its employees
the the the the income target right at
least the income target
by the way the principal business one of
the one of the
things that demarks principal business
is that it pays no higher than the final
income target so the final language
income target is about 107 000 token
tokens and dollars and uh a principal
business would never pay
someone more than that and ideally most
people who join a principal business
would become members and
i think they all would become members
and participate and delete it itself
so uh so that's that's a few things that
distinguish a principle of business and
obviously a community would want to fund
principal businesses because
this is a business model that is
designed to
support the lead and club model itself
so it's it's designed to support the
community and
if i were voting in the cbfs i would
really
you know like i value being a member i
value my community and i want my
i want this thing to work and i want i
want to receive the benefits that this

system is promising and the and the the
way to get there or one way to get there
is to mostly
fund principal businesses so that they
are serving the community in the way
kind of like a non-profit service a
community
and and they're paying wages that are
consistent with how this whole thing
works
and they're really being responsible to
the community
via this business model and it's really
in my best interest to support
principal businesses with you know
subsidies and donations and
loans and things like that not that
not that there's no room for for normal
for you know regular businesses
there is you can fund those too but you
know by and large you want to put
a little extra emphasis on principal
businesses
all right um
[Music]
yeah that's it for parameters next slide
now we're going to
start looking at what actually happens
um
uh we have about 20 and we have about 35
minutes left i think
we're going to get right through the end
of this so i may have to go a little
faster
as we as we get to the end so the first
slide on the upper left
is uh the fraction of the county
population and you can see in that blue

line that's the that's the
the rate at which people the
participation rate which is
decided in advance you know this is what
we expect to happen
and uh there it goes straight up
and then it reaches the plateau after 15
years so that's
after 15 years 90 percent of the county
population has joined the leader
and they've done so because their income
has gone up uh the
the if i were if i were if i were a
member of a citizen in the county and i
didn't have a job
well for sure i would want to join the
leader because i'm gonna get my income
is gonna go way up if i join the leader
relative to
not joining the leader so uh i account
here for every
not every unemployed person or not and
workforce person to
to to actually join the leaders so they
can be part of the you know receive
funding from the nurture arm of the cbfs
um down below that there's the income
target
the income target the blue line again
after 15 years it reaches a plateau of
about 110
000 tokens and dollars and then it
levels up
and and again i want to emphasize that
at that level
every community in the country could do
this this is not preclusive if
one community does it that doesn't mean

that others can't every community could
achieve these levels you know
based on the assumptions of the model
on the right is the fraction of income
uh over the years so
a fraction in the let's just look at the
the green line the green dotted line is
the token share of income target
so uh in the early years that's a very
low
fraction most of the income is in
dollars and then it raises over it rises
over the years to eventually reach about
35
so at the end of the simulation
about 35 percent of incomes and tokens
and the rest is in dollars
um and then the fund starts with the
bottom
right graph this is the average family
income
uh starts at about forty thousand that
is you know from this
from the census data and starts at about
forty thousand and you can see what
happens
with the blue line over the period of 15
years
it uh more than doubles
so family income more than doubles it
reaches the 90th percentile of starting
income
the red dotted line is showing the
similar thing only it's including the
money that people have in their savings
account
in the in the cbfs so there's a
two-person family each is contributed

to a maximum of 30 000 to the savings
account so
on top of this high income families also
have sixty thousand dollars in savings
in the in the cbfs
um you might notice that uh you know the
the
the all the growth is happening during
the first 15 years
but you can tell from the blue line the
family income line
at the bottom graph that it there's a
little bit of a lag it takes a little
bit of time for the whole thing to
actually reach
its maximum
family income and that happens about
your 28 or something like that
i might also mention too that the i've
just tried
in this simulation i was just trying to
find kind of reasonable
parameters and and a growth period and
things like that
such that it would all make sense and
would work and
we're not asking we're not expecting the
community to change on you know in year
one
suddenly everything changes it wouldn't
be like that it would tend to grow
slower
but having said that you could think of
the simulation as what might happen in
the first leta
but imagine that there's already a
hundred liters or 200 or 500 liters
around it's really clear how these

things work there's lots of data coming
in from leaders all over
it's clear that incomes are rising and
the
interest by the general population is
you know growing
very rapidly like suddenly it's
the new thing and everyone wants to join
because everyone's income goes up and
their
empowerment goes up and their sense of
community goes up and they're and
they're aligning their purpose with the
you know some greater good that's
it feels good you know there's a whole
variety of benefits that come from being
a member
so so now a new so someone wants to
start a club in
in another city and suddenly there's a
tremendous interest on day one
well then all of this would go a lot
faster it wouldn't necessarily take 15
years it might go
you know in half that time or whatever
uh on slide 16.
on the left is the unemployment rate as
i mentioned it starts at about seven
percent in the county
and you can see that uh after about
15 years uh the county achieves
full employment so uh that all works
well the lead is
generating jobs and whether they're
non-profit jobs in the nonprofit sector
or for-profit sector
the unemployment rate everyone's being
engaged that wants to be engaged now

keep in mind that a certain percent stay
out of the workforce and it just shows a
typical
percent whatever that was 36 or
something so that not everyone is
working some people are
doing other things in non-formal jobs or
whatever term you might want to use that
you know everyone's engaged somehow
people are actively engaged in their
in their lives and in their community
but some people
who are in the workforce the
employment rate is very low and and
the unemployment rate in the club itself
uh reaches its
lowest level after 10 years it just
takes a little while for the rest of the
county to
catch up on the left this is uh this is
the before and after
snapshot of what family incomes are uh
the upper graph shows the distribution
from the census data on day one or day
zero
and i uh the bottom graph shows the same
thing on
the 28th year of the simulation and you
can see that essentially
everyone in the county uh 90 of the
people in the county
are making the income target which is
just over a hundred thousand
so so it goes from a
highly unequal uh income
to a almost completely equal income over
the period of 28 years
let me let me ask a question john so i i

still see that between 350 and 400
there's still the uh 10 who don't join
uh ostensibly because they would stand
to yeah
right so how do you take that's great
they don't they wouldn't join a big
oh sorry just how do you take the other
90
and make their average value something
with fewer people working productive
jobs
having a higher average than 80 percent
that's just below 100k in the top graph
um if i understood the question it's
kind of what do you do with the night
with the 10 who don't join and do is
there some kind of do their income
somehow
equalize over time or do they like what
happens
well let me just ask again because they
they don't change so ignoring the ten
percent whose income does not change
how do you take the um belt the bell
curve
you know right side here which has an
average below 100 000
and then by reducing the variance also
increase the mean
while potentially having fewer people
with income generating jobs
just how does the extra usd
per year come into this system
okay again i'm not sure if i understand
the question but how does the extra usda
us dollars come into this system that's
through balancing of the trade deficit
so uh that's how that new dollars come

in that weren't there before
right so what the what this club is
doing essentially
is adjusting its trade balance with the
rest of the world
such that it's getting its fair share of
dollars
and uh you know the the dollar income of
the club
equals the average dollar income of the
of the us at the end of this simulation
did that answer your question uh
so it's predicated on basically
a locally positive economy that's
generating
a trade excess to get more usd because
you need
more usd to pay that number of people a
hundred thousand versus
between zero and a hundred thousand and
so even though fewer
are making potentially productive jobs
fewer are not in the workforce
there's so many more usd coming in that
we can raise the income for
them and everyone else i'm just saying
that that that's built in right
that's built in yeah that's all built in
yep yep
now what i thought you were originally
asking is what do you do with the other
10
what happens to them we'll talk about
that in a moment but i'm going to argue
that their incomes actually fall
towards this um this plateau of around
110 too
i think social pressure social pressure

would ensure that
that would be one that would be one
thing and there's some other reasons too
um well we might as well talk about it
now since we're on the topic
so uh so who's so so let's suppose that
i own this uh some
corporate i own a franchise of a
corporation i would say mcdonald's or
something could be anything
and uh i'm you know i take in money from
the local community they spend money to
buy my food but then i'm
sending money outside of the community
because i'm paying you know franchise
fees and other things corporate fees and
other things
so some of my money's leaving the
community i'm paying low-wage workers
and that
that's what makes this whole model work
is that i can get low-wage workers to
you know sit behind the counter and make
hamburgers or whatever
now what happens when there's no low
wage workers to choose from like
everyone's joined the leader everyone
who i would have paid
thirty thousand a year forty thousand a
year is now making a hundred and ten
thousand a year
where does my workforce come from how is
my business model going to even function
if i don't have a low low income pool to
draw from
so i i'm going to my business model is
going to be
strained to say the least uh it won't

work very well it's changing the game
right it's changing the foundations of
how the
normal business works in the world um
so and not only not only would people
not be available because i
because the leader would be paying
higher incomes
higher wages but but those people should
be enjoying their jobs a whole lot more
they would be
they would understand that they're
helping the
community by working in the bakery or
wherever they're working it's part of a
community effort
to raise the fitness of the community
right like we're all in this together
and it's and it's and i'm getting a
voice of
decision making by becoming a member so
there's lots of reasons i would want to
be a member and
very few reasons i would want to be a
low-paid worker at you know
at mcdonald's or anywhere else and
that's that's another reason and the
third reason is there's the ip
commons that i mentioned earlier um
so there's all these you know the leader
has now funded all these new businesses
some are technical businesses some
other businesses but there's all kinds
of property being generated
in particular intellectual property that
uh
that is shared amongst the membership
amongst the club and maybe even

between clubs so suddenly there's this
enormous advantage to
[Music]
to principal business or to uh to member
businesses to becoming members because
you get this you know this volume of
a free intellectual property and
standard businesses that are outside of
this do not have access to
that to you know they have to pay pay
royalties to get that
you know ip stuff so that's another
reason
uh and i think there's actually still
one more reason that i can that i
had i want to talk about
well a successful community
a successful community would have many
leverage points
and many ways to draw income in and an
unsuccessful community would have
no one wanting to help them except for
the us government paying usd which i
also saw in the model right that the usd
is providing
by the government so it's like welfare
at the country level
and also at the community level yeah
yeah so imagine that's a good point and
imagine how that's
in fact say a county government
county level governance or state level
governance or
even federal governance what if it
didn't have to pay
like all this unemployment insurance and
all all these other things
fees that the government has to pay you

know when people are
when people need help right all these
programs
and the and all of it is being taken
care of at the community level by
leaders and clubs like leaders so the so
there would be actually like in that
sense you would
expect that the local state and gov and
national governments
might actually like this concept you
know because it's reducing
their liabilities you know that's
especially at the local government the
local government as i said the local tax
revenues
are double in this in this simulation so
you know
the county government might go like hey
good idea
this is if this could work this is a
great idea and the
individuals might say hey this works
this is a great idea but my income is
going to double
uh great fantastic uh just a bit on
an income obviously with this the lower
your starting income
the more you benefit from this so if you
happen to be
someone at day zero where your family
income is fairly low
yours is going to have the greatest rise
compared to someone who's already making
a
say mid-level income or something but it
doesn't matter it is like the rising
boat catches everybody by the time it

gets up to them
up to the plateau after 15 years
uh some more results on slide 17.
uh on the upper left hand corner uh
you can see how much money is
circulating in this uh
cbfs and um
i i want to emphasize again again that
this is a community
of a hundred thousand people this is a
small community right so this small
community
is now circulating six billion in
currency
annually that's six billion that it has
to choose
about like what jobs do we create what
what what nonprofits do we
support what do we want what kind of
parks do we want
what is the road quality how's our
hospital system
whatever decisions this community needs
to make it's got six billion in currency
to play with
so uh an enormous amount and that should
be enough for the community to
make whatever choices it wants as far as
like what are we going to do how are we
going to get there
are we going to improve our quality of
life
and maybe just on that note i will
mention again we've already said this
but
there's no club that's a that's an
entity even
in and of itself as clubs start to form

and
uh and develop there would also be along
with them
networks of clubs that are forming and
developing that are also
a cognitive system so the cognitive
system is playing that
cognitive architecture is playing out at
the local level at a club
and it's playing out between clubs as
clubs need to make decisions
about how they're going to solve larger
problems right
so so again no club is in this for
its own good and only its own good is
this is a you know clubs are connected
this is a
greater good project um
the bottom left these are how the
workforce gets divided over time so on
day zero
most of the workforce is in the regular
state of businesses
almost 90 97 is in a regular standard
business a little bit is in the
nonprofit sector and obviously zero is
in the principal business sector
but after after the period of 28 years
it achieved some
different balance of the torque force
and i just want to say again that
whatever balance that is it's a balance
that the community chooses in advance
right so this just didn't play out by
itself these are based these are all
based on community decisions
all along the way like we want this
community wants to achieve

a fifty percent uh you know that to
split its workforce
uh fifty percent into the nonprofit
sector or forty percent or eighty
percent or ninety percent or whatever
it wants um upper right draft
are the donations made to nonprofits
these are dollar donations apart from
the cvfs
and i'm just using a like a fraction of
income goes to donations like a general
fraction
and you can see that uh donations to
local nonprofits more than double
so you know this at least in theory this
whole system
is like a boom for county governments
it's a boon for the non-profit sector
it's a boon for the education sector uh
and then everyone else's income is going
up and unemployment's going down too so
like who loses the only people that are
losing from this whole thing
are the people who hold enormous
economic power
on day zero you know like those people
aren't really gaining from this
and if lots and lots of clubs were to
you know lots of lots of cities were to
implement clubs and this actually
happened
in scale you know distributed at scale
then the real losers in a sense would be
those who are
quite you know quite wealthy to begin
with like billionaires say or
multi-millionaires because their income
their incomes would probably drop as

i've mentioned
over time only because of social
pressures and because they can't get
people to work in their businesses and
things like that
but at the same time everyone's
well-being increases so it's not like
anyone loses everyone
wins here like the commute the world is
you know all communities are
more empowered to solve their real
problems that matter this is a cognitive
system
aimed at achieving sustainability and
thriving
into the future so there really isn't
anybody that loses from this
there's just some apparent losers you
know and not very many of them
at that um the
lower graph on this i won't say too much
i'll just
explain the blue line there this is the
number of tokens
uh that are generated by the community
so obviously in the early years you're
for the first 15 years you're generating
tokens
uh because you're growing in this growth
period and then after that you have
the tokens are just circulating and
importantly they don't circulate outside
of the membership so if i'm a standard
business
i there's no one can oh that's the other
reason why why
that my clubs would and club businesses
would

grow is that if you're not a part of the
club people can't spend their tokens
you know at your establishment so i i
might
be a local business and i don't want to
participate and that's fine it all goes
as it goes
but everyone else's incomes are
increasing part of that income isn't
tokens and those people
can't spend their tokens at my
establishment the only way they can is
if i join the
if i join the club then they can spend
their tokens at my establishment
uh page 18 and i think this is the last
of the results i was going to show
this is the results just for the local
government this is tax revenues for the
local government
at the red is the tax receipts so for
this
for this uh lane county is estimated
that it starts at about
a billion dollars annually in revenues
and it more than doubles
by the end of the simulation so their
tax revenues go way up
and you can imagine what counties would
want to do with
you know if their revenues tax revenues
doubled
you can imagine they could put that to
very good use
okay then just some final remarks on
these last two slides
um and this this first one is something
that we

just were touching on um what happens to
the other 90
that don't join the leader right so i'm
i would argue
that in the long run even that 10 that
has very high incomes would see their
incomes
move closer to the to the income of the
target income for the leaders and that's
for all the reasons that we just
mentioned
that that uh
business owners don't receive the
benefit of tokens because they can't
accept tokens
uh there's social pressures to belong to
the club because it
it's in everyone's benefit for people to
belong to the club
uh the ip is there's an eye by people
that people have access to if they join
the club
and so like a number of reasons maybe
there was one other that we talked about
uh so you would you would think that
you know this is the new you know like
if this is happening that it's the it's
it's like who who uses a typewriter
anymore right
hardly anybody but it's because
computers work better for
type setting you know this is technology
that's another way to say that this is a
technology this is a decision making
technology
and it's both a soft and a hard
technology
education is part of it there's you know

there's all these components to it but
it's technology
and uh you know that's what happens in
the world a new technology springs up
it's it's far better than the old one
and then suddenly
the old one is is toast well
the old one uh becomes subsumed into
conflict at a higher level so you could
have different
uh chains that are competing so uh the
idea that
still one part for me is that
just thinking about evolutionary
transitions in individuality it raises
the level of conflict to another level
up from what we've seen before
with everything that politically and
game theoretically would come with that
and the second piece is um um i i don't
update my priors
based upon models without parameter
sweeps
so it's hard to look at a given layout
and imagine
but it's something that we actually use
as a north star
for people who are on that political
journey they might see that as
beneficial
but as you're pointing out people who
don't see it as beneficial are not going
to join
until the incentive structures are
restructured for them
or that will happen yeah yeah absolutely
yeah
and everyone's learning from this so you

know the communities the world is
learning from this right
and the world is learning over time
whether this is even a good idea or not
does this whole system work
and obviously if it doesn't work if it
doesn't produce these benefits that are
that are that are expected in advance if
it doesn't do
that then it doesn't work and then no
one would join because it's a stupid
system
right or it has flaws that need to be
fixed or something
so uh yeah everyone's learning together
and um
and uh there could be numerous trials
going on of different systems at the
same time and you know try for the world
everyone can be trying to figure out
what works best in this what kind of
systems work best
and um yeah all of that's happening it's
evolving
as as we go it's evolving but the
important thing is
is that this is a framework to
facilitate societal cognition you know
at the local level
and um if we think of our
societal systems as cognitive systems
then it's a
perfectly reasonable question to ask
what kinds of system designs work best
and that's what this whole r d program
is aimed at just
asking the question of all conceivable
designs

which kinds of designs might most
facilitate
societal cognition and therefore achieve
the intrinsic purpose of a of a society
which is to
thrive now and in the future so what
designs
and you can imagine all kinds of trials
to get to help answer that question
um next slide and this is the last one
just a few remarks i've touched on this
already but
but an organism is agile right like i
have
among among other things i have legs i
can run from danger i can do things i
can change my behavior
uh as the community we can we can change
our behaviors we can we can do whatever
we need to do to achieve whatever
task is in front of us or you know at
least achieve most tasks
but in particular elita is agile like it
is flexible it can choose what jobs it
funds and suppose that you know 10 years
into a
into a new club some big thing happens
or some new challenge that wasn't
anticipated it just arose
well this is decision making under
uncertainty that's the whole concept of
this
like this whole system facilitates
decision making under uncertainty
something new has happened now it has to
the community has to respond to
this new set of conditions it can change
its makeup of

what businesses it funds what
organizations it funds who does what
it can act it you know it can change the
how many people work in the nonprofit
sector versus for-profit sector
it can essentially use it can vote to
to focus its attention and its resources
human and otherwise on whatever problem
it needs to focus its attention on so
it's agile by design
and part of that by the way is is having
a fairly equal income so in our native
system
if i'm a billionaire it is really hard
for the
society to allocate my billions of
dollars for some worthwhile project
right because
because of the you know the private
property laws that's my private property
and you can't use it right
but if everyone's income is equal then
then everyone is participating in that
redistribution of attention
and resources to whatever problem is
important to the community
so uh a community empowers businesses
with with funding for example and it's
important that it can retract
that uh support when necessary
so the the design of this whole system
is such that
there that things can change uh rapidly
if need be and the community can
redirect its attention to whatever
problem it needs to address
um again the next one there is that the
main resource is not money it's human

it's human attention and and energy and
also to some degree
natural resources but the real purpose
of money is simply to
to allocate human attention and human
energy and resources into those problems
that actually matter
and and lastly again uh the leader can
because of its agility because members
are choosing what kinds of jobs to make
what
what sorts of businesses what sorts of
right what sorts of
economy it wants it can choose any
flavor of an economy it wants it creates
the jobs that it wants to work in and
the kinds of jobs that the people want
their children to work in
so it can do whatever it wants as long
as it functions as a cognitive system
and as long as it actually meets the
essential needs
of the organism of the community and
that is uh
it in a nutshell or not in a nutshell
that's it in those
long distances there's the the blueprint
haiku for the nutcracker
it is a few levels upstream but that's
why it's fun to
have these open-ended discussions and
just allow um
what you've worked on for so long to
flow out just hear how you're laying it
out
and maybe in closing out this little
discussion as we head into the next two
weeks with the

group discussions what how would
somebody get involved
or where can somebody come into play
after seeing this or reading the papers
oh well um i can be reached through the
website principal societiesproject.org
and there's
there's other resources there so you
know one start would be just to
familiarize yourself
with the material a bit but we have a
mailing list
so people can just join the mailing list
and as things
progress and as opportunities arise then
we can shoot you know we can use our
mailing list to contact people
um this our this whole series is written
to the scientific community
so that's that's my target audience for
the series of papers and i'm hoping that
i'm hoping that i can entice the
scientific community to pick up this
topic as a worthwhile topic of
study and endeavor you know can we can
we as a as a scientific community
address these issues
create designs test designs identify
fitness metrics for community you know
describe how community cognition works
and what is how it works best and the
conditions you know necessary are we
are we ready to to pick up this topic
and i'm hoping the answer is yes
because it looks rich to me and on so
many levels could be extremely valuable
to society it
and obviously it's aimed at societal

transformation
and it's already quite clear that that
different groups
academic and otherwise generally
understand that bold change is necessary
so this is
one form of bold change now uh academics
in particular please do reach out if you
think if there's any number if you have
some funding
there's any number of small studies that
could be done right you know immediately
this simulation could be uh um
expanded in a number of ways uh a study
could be done
just looking at just asking the public
what they think of deep transformation
and what they would think of of
income equality or what that you know
like laying out some
possibilities to the public or if you're
in media
whether that's literature or music or
movies or documentaries or whatever
please contact me because
just getting the word out to the public
that transformation in this sense
might actually be viable and possible
that that's like half the battle
is just to say we can do this there's a
way we can do this this is open to us if
we choose to take it and uh
so that that's important there's so if
you got any idea
about how to what you'd like an idea of
how to contribute to this
effort then great you know send me note
it's like either they can read it now

and send you a note
or hear about it later or not so
that's where it goes john
thanks so much for these awesome
sessions though because it was great
reading the papers and great to go
through them again
and we'll all be looking forward to
speaking with you
on the upcoming tuesdays at 7 a.m
pacific
so we'll talk to you soon great i really
appreciate the opportunity absolutely
john
talk to